



Automatic Toy Sorter

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Introduction

Every parent has experienced a time with their young children in which cleanup and organization after a long day of play is a seemingly impossible task to achieve regularly. Children naturally experience time blindness, a phenomenon in which they lose track of time and get distracted by what the world has to offer. This is a problem that can be easily solved through technical software and robotics. OpenMANIPULATOR-X is a robotic arm built to make the clean-up time for parents and children a little less tedious. Its design allows for five degrees of freedom, enabling it to sort toys and items based on AR Markers automatically. It includes advanced features such as x,y, and z, enabling it to be the perfect tool for that after-play clean-up.

Project Goals

Problem Statement: Cleaning up and sorting objects is time consuming, and can limit efficiency. During a busy day, cleaning up objects isn't a first priority, which results in an inefficient and inappropriate working environment. For kids, it will be harder to walk inside their room, and they will lose more objects.

Constraints: The width of the objects being picked up must be within 2 centimeters. Along with that, the robotic arm (OpenManipulator-X) can only pickup objects that weights under 500 grams. Each motors on the arm can only move a certain degree range, and the objects must be placed in a specific range, so the camera can detect and grab them from an appropriate distance.

Solution: Implement OpenCV, and ArUco Marker detection. All the objects will have an AR Marker, and they will be picked up by the robot, and placed in the containers, corresponding to the AR Marker attached on the object. A simple velcro

Overview of the Required Components

- LogiTech C920** : By utilizing OpenCV, the camera will detect objects that will have an AR Marker attached. Alongside, it will also read the AR Markers on the containers, to drop them accordingly.
- Dynamixel X430**: The motors that is on the robotic arm, OpenManipulator-X. The arm will have five degrees of freedom, and will be programmed to pickup the objects and drop them in the appropriate container.
- Containers**: Used for sorting. The containers will store the objects, and can easily be picked up all together and stored somewhere for later purposes.
- AR Markers**: Will be attached to all the containers and objects that will be picked up. The object's purpose is to be stored as a name, and will be dropped to the container that has the same AR Marker (or name).

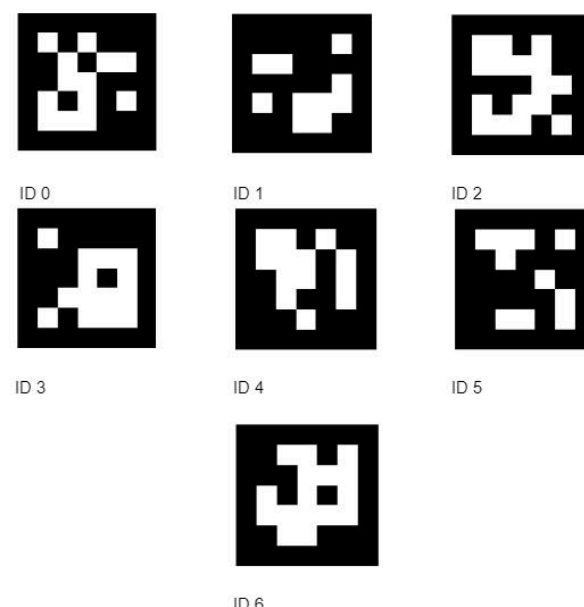


Figure 1b: The ArUco Markers used for this project. Will be 5X5, standard directory



Figure 1c: LogiTech C290 on the robotic arm

Graphic User Interface: Arm Application

- Arm Movement controls**: The GUI takes custom coordinate inputs and directs the robotic arm to the coordinates inputted. Alternatively, arrow buttons tied to keyboard arrow keys will move the arm in those directions.
- AR marker list with toy assignment**: For users to customize the auto sorting. This menu keeps track of seen AR marker ids. After selecting the id, the user may assign detected toys or from a list all detectable toys to it.
- Save/Load**: The program saves all toy-to-markerID assignments and loads them into the program every boot up.

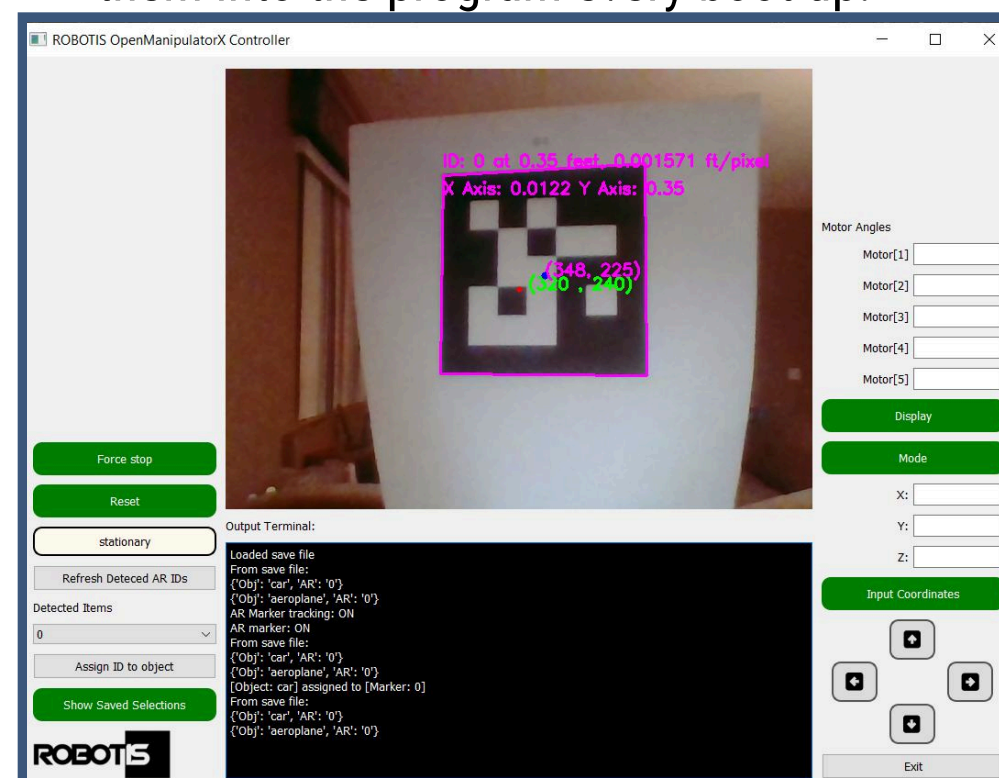


Figure 2a: Controller GUI window

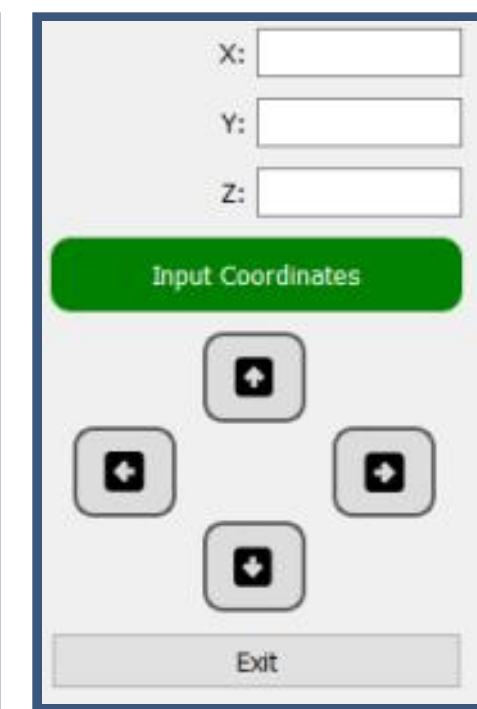


Figure 2b: Manual Arm controls

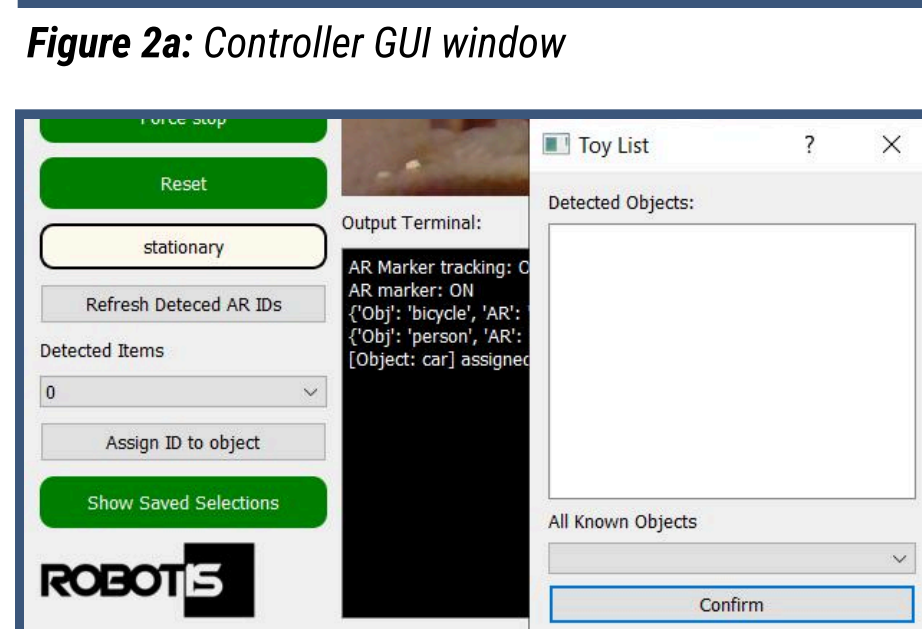


Figure 2c: Refreshing menu saving detected AR marker ids and object assignment

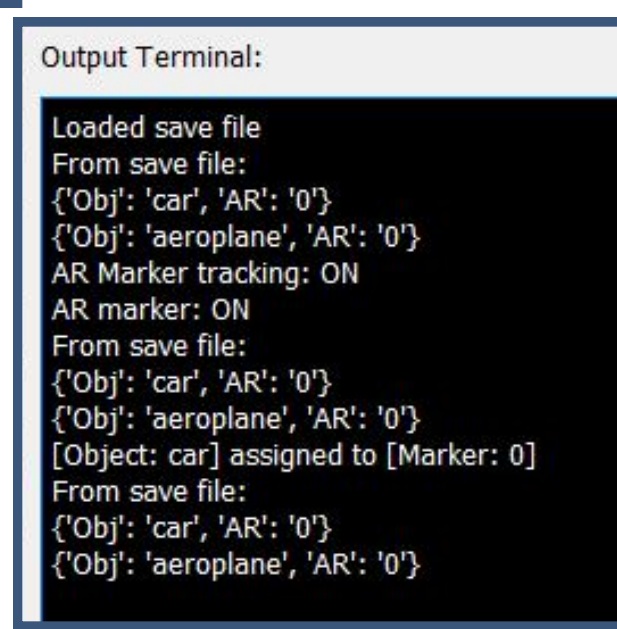


Figure 2d: Output Terminal displaying tasks completed and text feedback

ArUco Markers and Object Detections (OpenCV)

ArUco Markers: Attached on each containers. Containers will have a set of poles (2 of them) and a small square raised, for the LogiTech C920 camera to detect. Once it detects the appropriate ArUco Marker to go, it will release the toy from the arm, and drop it inside the container, after appropriate movement has been made.

By using the LogiTech C920 camera on the OpenManipulator-X, we can import cv2, from OpenCV. Afterwards, there are 2 methods to sort the objects



Figure 3a: Picture of the container

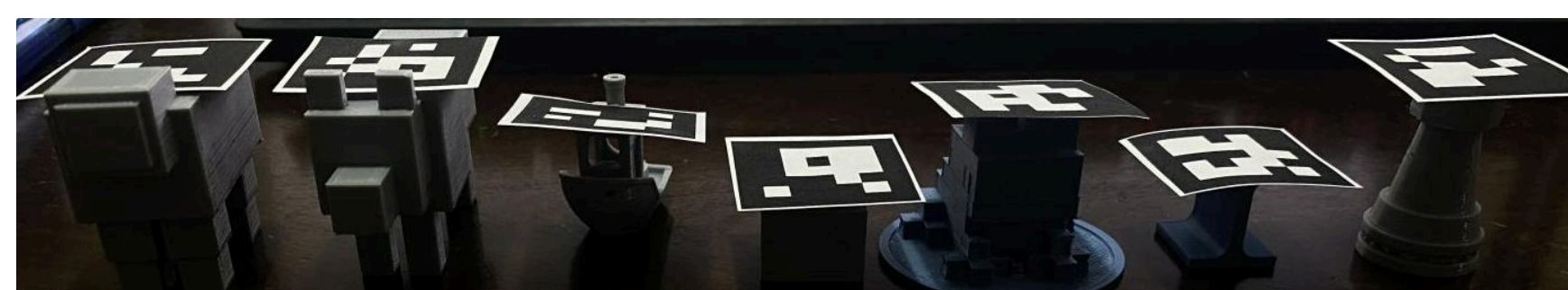


Figure 3b: ArUco Markers placed on top of the toys, before implimenting YOLOv8

User Experience and Customization

- There's a help menu on the task bar in the user interface that describes the purpose of each function.
- Display button opens a menu for enabling/disabling the AR marker and object borders on the video feed

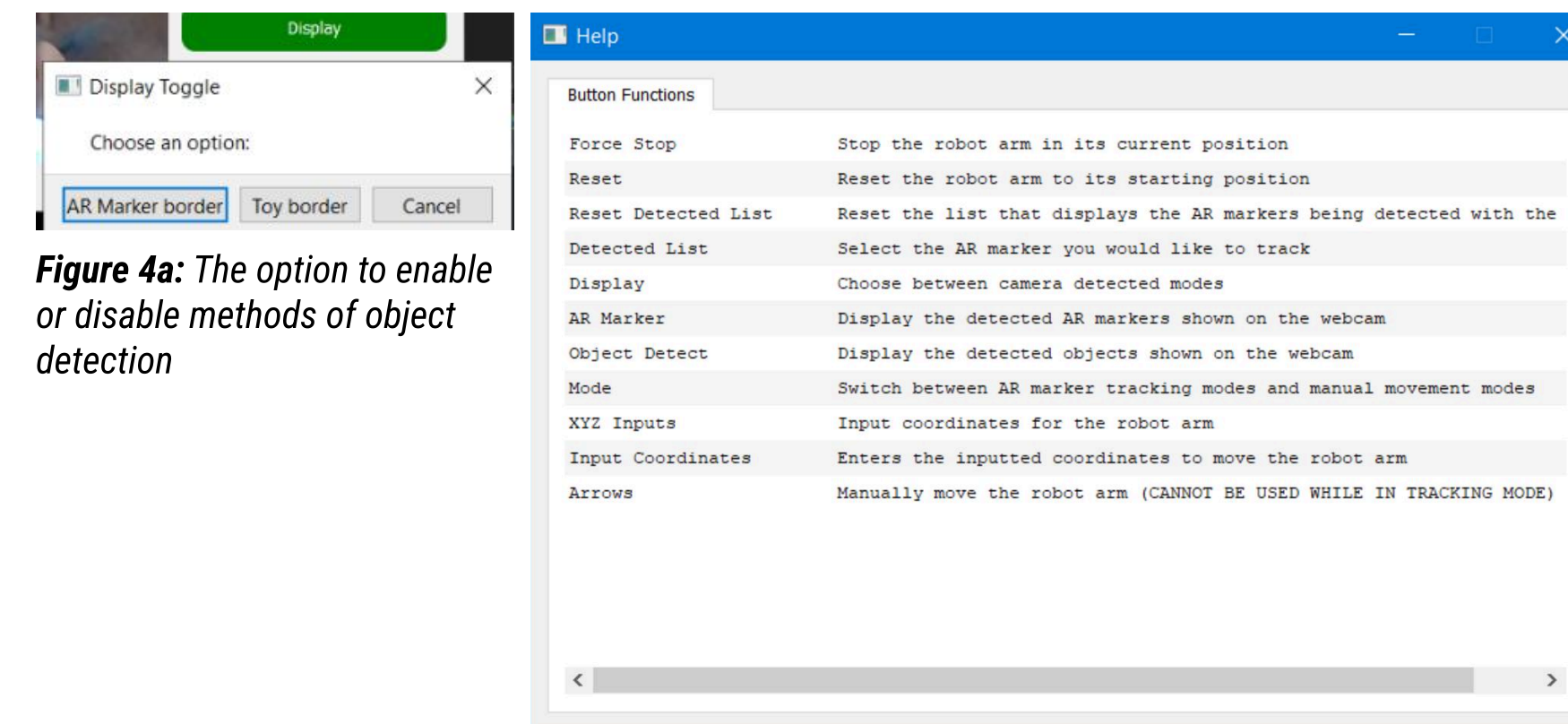


Figure 4a: The option to enable or disable methods of object detection

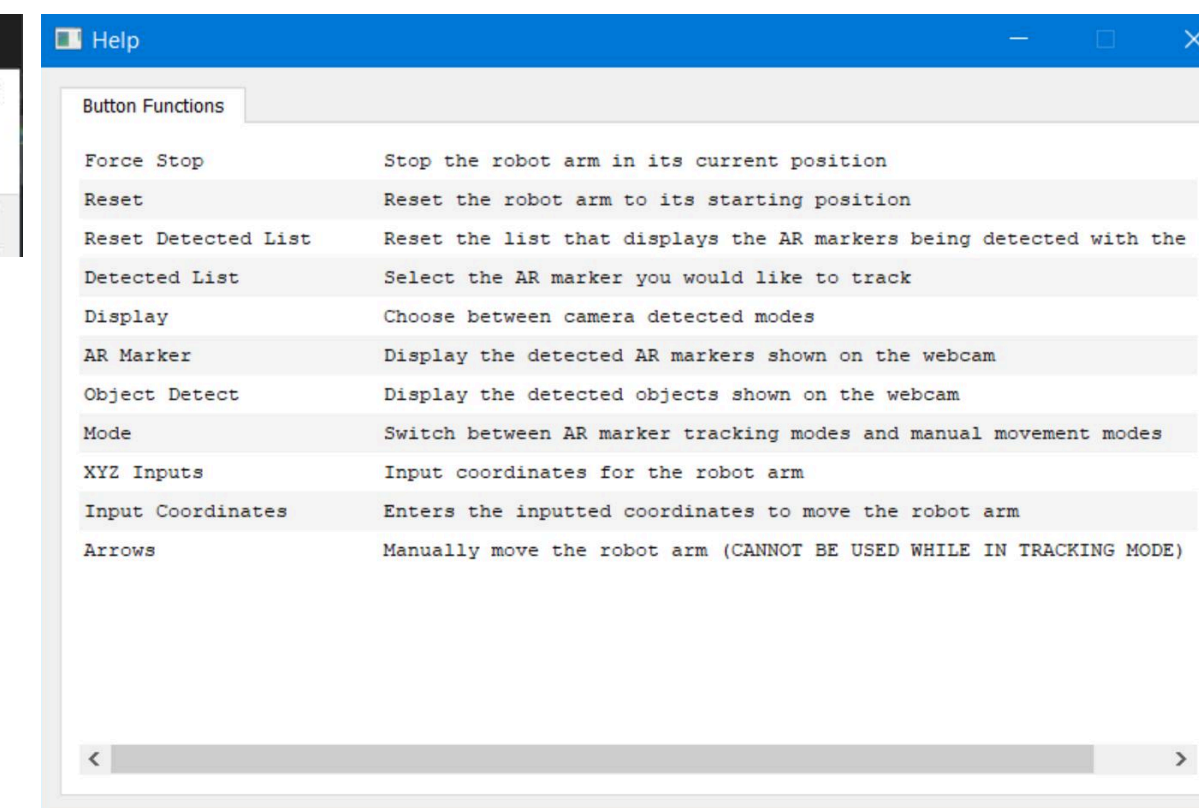


Figure 4b: Help menu located on a drop down menu at the top of the user interface

OpenManipulator-X Application and Limitations

- The robotic arm (OpenManipulator-X) is used to pickup objects.
- The 5 degrees of freedom allows the robot to move up, down, left, and right. As a result, there are 3 coordinates that should be considered
- Adjusting angles of each servo motor will help make sure the arm will move
- The robotic arm has a payload of up to 500 grams. The toys must be lighter
 - Will use a 10% fill for all the toys 3D printed
- The grip can be a maximum of 7.37 centimeters
 - The toy's width will be 2 centimeters
 - The toys must be positioned correctly. If the length is positioned to get picked up, it's going to fail

Servo ID number	Purpose
0	Gripping the object
1	Moving left/right
2	Moving up/down
3	Moving up/down and forward/backwards
4	Moving up/down and forward/backwards

Figure 5: The servo ID number, and it's purpose

Inverse Kinematics Calculations for Z-axis Arm Drop

Angle Ranges of θ_w θ_e θ_s

$$\theta_w : [-90^\circ, 90^\circ]$$
$$\theta_e : [0^\circ, 207^\circ]$$
$$\theta_s : [-30^\circ, 200^\circ]$$
$$Z_T = l_w \sin(\theta_w - 180^\circ) + l_e \sin(\theta_e - 47^\circ) + l_s \sin(\theta_s - 270^\circ)$$

$w = \text{wrist}$ $e = \text{elbow}$ $s = \text{shoulder}$

Figure 6: Calculating the total Z-axis required by referencing the displacement of the claw regarding the arm base.

Object Sort System Diagram

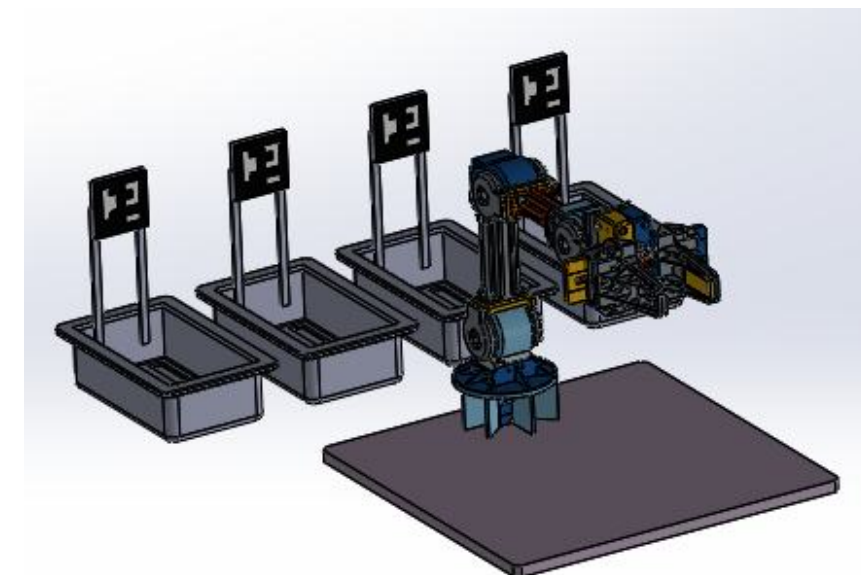
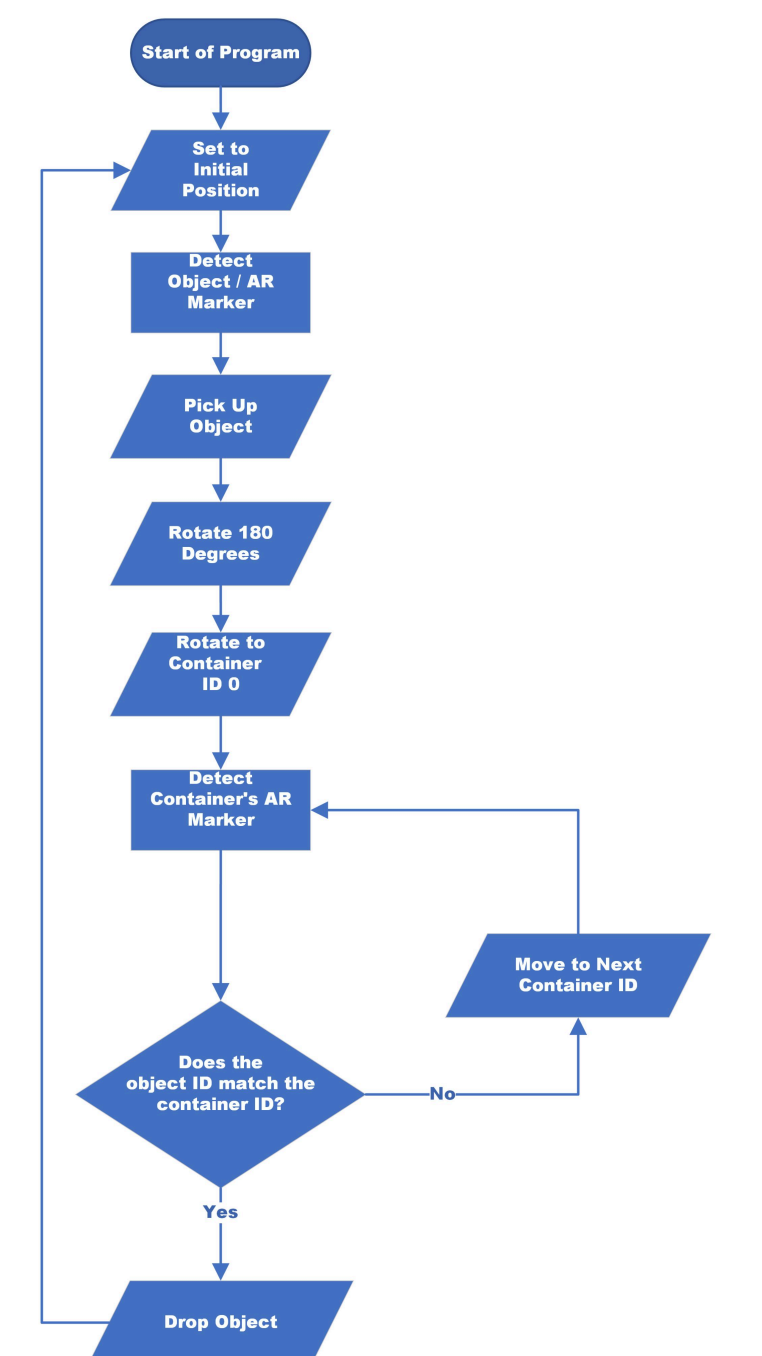


Figure 7a: Automatic Toy Sorter System. Toys will be placed on the aluminum platform.

Figure 7b: Flowchart of the Toy Sorter System. Shown as a loop from setting to initial position to dropping the object.



Budget

ROBOTIS x BANSHEE UAV Internship 2023 - Bill of Materials					
Type	Manufacturer	Part Name	Price Per Order	Quantity	Net Cost
Battery	Ebay	Zee 4S 5200mAh Lipo Battery	\$50.00	1	\$50.00
Battery Charger	Ebay	LiPo Charger	\$30.00	1	\$30.00
3D Printed Poles	Newtonnate	3D Printed Poles	\$0.16	8	\$1.25
3D Printed Squares	Newtonnate	3D Printed Squares	\$0.99	4	\$3.94
3D Printed Cubes	Newtonnate	3D Printed Cubes	\$1.54	4	\$6.17
3D Printed Toys	Newtonnate	All 3D Printed Toys	\$3.42	1	\$3.42
Motor	ROBOTIS	Dynamixel X430	\$289.00	1	\$289.00
Camera	Logitech	LogiTech C290	\$56.67	1	\$56.67
Robotic Arm	ROBOTIS	OpenManipulator-X	\$1,416.36	1	\$1,416.36
Containers	Walmart	Containers	\$2.49	4	\$9.96
Subtotal					\$1,866.77
Tax (-10%)					10.00%
Grand Total					\$2,053.45

Figure 8: Table containing Bill of Materials and Total Budget budget.

Conclusion and Future Works

This project provides a method to pick up toys that can meet the restrictions (2 centimeters), and moving it to the appropriate position before dropping the toy inside the container. Utilizing OpenCV, and AR Marker Detection will help the toys to be placed appropriately, and picked up accordingly. As the toys are being picked up, it provides a lot of time being saved for the kid or parent. While this automatic sorter sorts the toys, they can do other things to be more productive with their time, such as errands, working on important documents, or doing other things that can benefit them

Using machine learning to train and detect toys without the need for physical identification beyond the object itself. The project will make an attempt to add an extra degree of freedom to twist the claw of the arm sideways to pick up toys that are tricky to grab onto straight on. There will also be an extra flexibility to types of grabable items by reworking claw design.

Acknowledgments

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